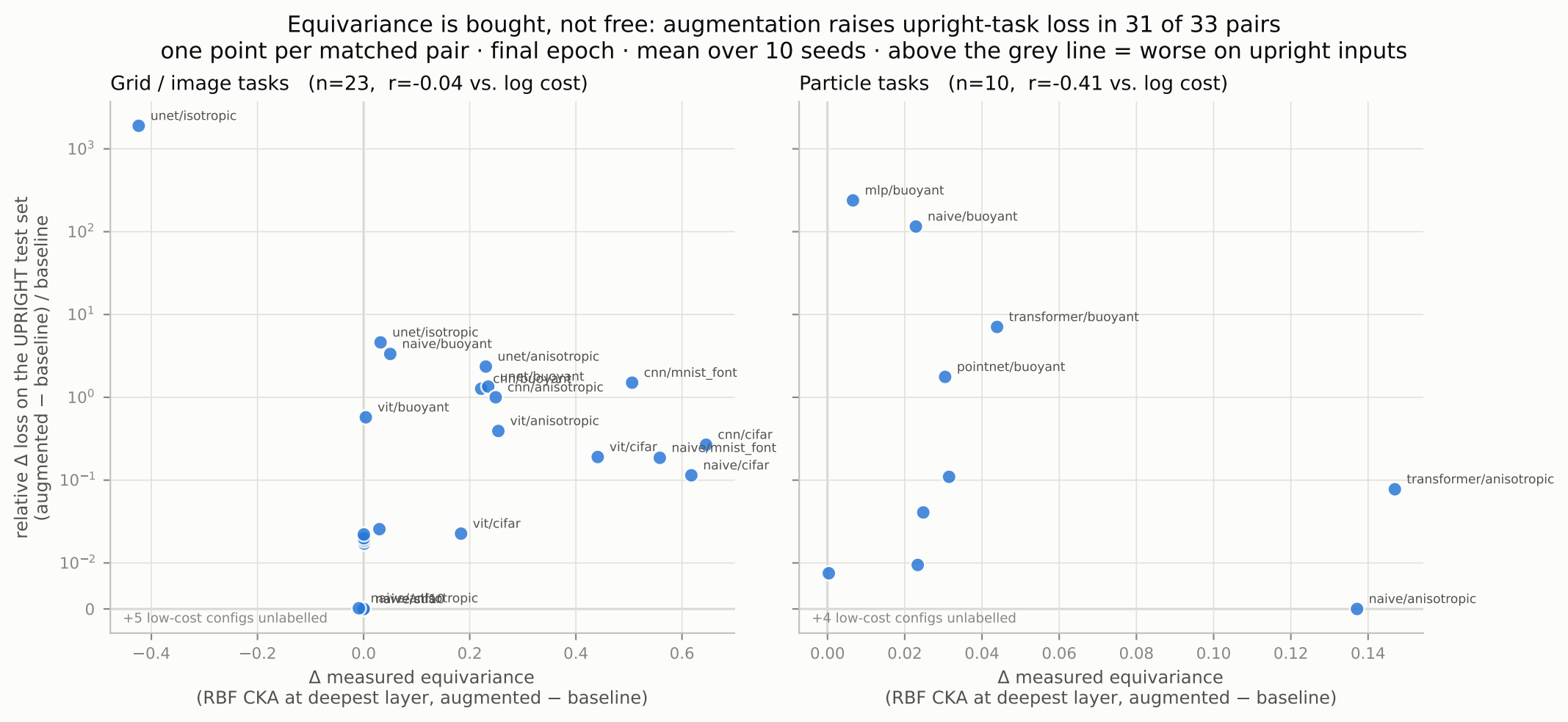

Equivariance is bought, not free: augmentation raises upright-task loss in 31 of 33 pairs
one point per matched pair · final epoch · mean over 10 seeds · above the grey line = worse on upright inputs
Grid / image tasks (n=23, r=-0.04 vs. log cost)
Particle tasks (n=10, r=-0.41 vs. log cost)
relative Δ loss on the UPRIGHT test set
(augmented − baseline) / baseline
Δ measured equivariance
(RBF CKA at deepest layer, augmented − baseline)
unet/isotropic
naive/buoyant
unet/anisotropic
cnn/mnist_font
cnn/anisotropic
vit/buoyant
vit/anisotropic
cnn/cifar
vit/cifar
naive/mnist_font
naive/cifar
naive/anisotropic
+5 low-cost configs unlabelled
mlp/buoyant
transformer/buoyant
pointnet/buoyant
transformer/anisotropic
+4 low-cost configs unlabelled